Aging reduces shared event structure at encoding, weakening memory reinstatement in the Posterior Medial Network

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ABSTRACT

Episodic memory involves forming coherent event representations during ongoing experience and reinstating those representations during recall. Although aging is thought to reduce memory specificity, it remains unclear whether age-related reductions in neural reinstatement reflect degraded event representations at encoding, impaired reactivation at retrieval, or both. Here, we used fMRI to measure activity in the posterior medial network (PMN) while younger and older adults watched a 26-minute television episode and later freely recalled it. We assessed three complementary measures: Within-participant pattern reinstatement, within- and cross- age group intersubject correlation (ISC) to measure shared temporal dynamics across individuals, and event-level intersubject pattern similarity (ISPS) to measure shared spatial representations across individuals. ISPS was computed for encoding-to-encoding, encoding-to-recall, and recall-to-recall event pairs, and was evaluated within- and cross-age groups. Older adults recalled the episode less specifically and showed weaker event-specific reinstatement than younger adults. During encoding, older adults exhibited reduced ISC and ISPS within their age group, while cross-age group comparisons showed no differences. Across both age groups, encoding-to-encoding ISPS was the most consistent predictor of reinstatement. In younger adults, reinstatement was additionally predicted by ISC and encoding-to-recall ISPS. Encoding-to-recall ISPS among older adults revealed a suggestive dissociation preferring alignment with younger adults' encoding patterns over those of other older adults. This dissociation occurred even though encoding-to-recall and recall-to-recall ISPS were comparable across age groups, arguing against a retrieval-only account of age-related memory decline. Instead, the stability and shared structure of event representations during encoding may constrain memory reinstatement, shifting the focus toward encoding as a key contributor to age-related memory change.

Significance Statement. Older adults often remember the gist of experiences but struggle with specific details. We show that this may reflect changes in how the brain processes events as they unfold. When younger and older adults watched the same television episode, older adults' brain activity was less synchronized with others and showed greater individual variability within their age group. Critically, older adults with the strongest memory reinstatement showed encoding patterns that converged on shared event structure, with suggestive evidence that alignment with younger adults' encoding patterns was especially important for bridging encoding and recall. This suggests successful aging may depend on preserving broader ways of representing events rather than following age-typical processing. Our findings reveal that memory decline reflects inconsistent encoding across individuals and not retrieval failure alone. It also suggests new avenues for detecting risk and improving memory as we age.

Keywords: *Episodic memory, aging, naturalistic, intersubject correlation, intersubject pattern similarity, pattern reinstatement*

INTRODUCTION

Our ability to remember rich, detailed experiences changes with age. Episodic memory, or the capacity to encode and retrieve specific events situated in time and place, declines even in healthy older adults (Craik & Salthouse, 2011; Nyberg et al., 2012). Prior work has shown that older adults tend to retain the high-level meaning of past experiences but lose access to contextual details such as precise locations, temporal sequences, or perceptual features (Schacter et al., 1997; Koustaal & Schacter, 1997; Grilli & Sheldon, 2022). For instance, remembering a dinner with friends may involve re-experiencing the flow of conversation, the layout of the restaurant, and the emotions of the evening, but older adults increasingly rely on the general gist of "a nice dinner out" while the specific moment-to-moment dynamics become harder to retrieve (Delarazan et al., 2025). These deficits may reflect difficulties among older adults in reactivating internal representations of past experiences, which are needed to reconstruct and describe events in the present (Nyberg et al., 2012; McDonough et al., 2014; St-Laurent et al., 2014).

Retrieving detailed event memories often relies on interactions among distributed brain networks, including the hippocampus, medial temporal lobe regions, as well as lateral parietal and medial prefrontal cortices that form large-scale networks (Buckner et al., 2008; Eichenbaum, 2017; Rugg & Vilberg, 2013; Wagner et al., 2005). Within this broader system, a posterior medial network (PMN) of brain areas, including the posterior cingulate cortex, precuneus, and lateral parietal regions, has been implicated in representing the structure of events and maintaining situation models that integrate information over extended timescales (Ranganath & Ritchey, 2012; Ritchey & Cooper, 2020; Hasson et al., 2015). PMN activity tracks changes in narrative and spatial context and is engaged when people retrieve, imagine, or update complex events (Baldassano et al., 2017; Benoit & Schacter, 2015; Chen et al., 2017; Robin, 2018), suggesting that it plays a

central role in organizing ongoing experience into coherent event representations and supporting their later reconstruction.

A key mechanism thought to support memory retrieval is neural reinstatement, the reactivation of neural patterns during retrieval that were present during encoding (Bird et al., 2015; Bonnici, 2016; Chen et al., 2017; Oedekoven et al., 2017). Work in younger adults indicates that naturalistic movie viewing evokes event-specific patterns of PMN activity during encoding, and that these patterns are reinstated during subsequent free recall of the same scenes (Chen et al., 2017l Oedekoven et al., 2017). Importantly, stronger PMN reinstatement is associated with better memory for naturalistic events (Nguyen et al., 2019, Danker & Anderson, 2010). Prior work suggests that older adults show reduced neural reinstatement during retrieval (Koen & Yonelinas, 2014; Campbell et al., 2013), but these studies have used controlled laboratory paradigms that may not capture how aging affects memory for complex, real-world events that unfold over time. Thus, it remains unclear whether older adults show comparable reinstatement deficits during spontaneous recall of naturalistic experiences, and whether any deficit reflects degraded event representations at encoding, impaired reactivation at retrieval, or some combination of the two.

Given that pattern reinstatement intrinsically depends on both encoding and retrieval, reduced similarity between encoding and recall patterns could simply result from noisier or less stable event representations at encoding. One mechanism that may shape the stability of encoded event representations is how ongoing experience is segmented into discrete events (Zacks et al., 2007). Aging has been linked to alterations in event segmentation signals in hippocampus and PMN (Reagh et al., 2020), which may contribute to more variable interpretations of an ongoing experience across older adults. Thus, it is useful to consider measures that reflect the reliability and consistency with which event structure is expressed across individuals.

One measure that directly assesses shared encoding dynamics is intersubject correlation (ISC), which measures the degree to which moment-to-moment fluctuations in neural activity are synchronized across individuals who receive the same time-locked input (Hasson et al., 2004; Nastase et al., 2019). In younger adults, higher ISC has been linked to better comprehension and memory of the stimulus, suggesting that tightly time-locked responses reflect shared, reliable processing of the experience (Hasson et al., 2008; Schmälzle et al., 2014; Su et al., 2024). In contrast, older adults show reduced ISC in PMN and frontoparietal regions during naturalistic viewing, and these reductions have been associated with poorer performance on cognitive measures related to attention and memory (Campbell et al., 2015; Geerligs et al., 2018; Cohen et al., 2018). These findings suggest that older adults may encode continuous events in a less consistent time-locked manner compared to younger adults, potentially constraining later reinstatement.

Beyond temporal synchronization, the spatial consistency of event representations can be assessed using intersubject pattern similarity (ISPS). ISPS measures how similarly different individuals represent the same event in terms of spatial patterns of neural activity (Nastase et al., 2019; Wang et al., 2020). During encoding, ISPS can be used to test whether an event elicits a shared neural pattern across viewers. In younger adults, these event-level patterns are not only shared during encoding but can also be recovered during recall and show convergence across individuals as events are reconstructed, consistent with a shared neural code for event representations (Chen et al., 2017; Zadbood et al., 2017). Higher ISPS also predicts greater overlap in what different people remember, suggesting that shared spatial patterns of neural signals capture essential pieces of a complex experience, which in turn can support robust encoding and later reconstruction (Koch et al., 2020; Sheng et al., 2023).

Taken together, intersubject measures allow us to characterize the extent to which event representations are shared across individuals during encoding and recall. It is possible that shared time-locked dynamics (ISC) and shared spatial event patterns (ISPS) support effective event processing, such that greater convergence with these shared responses is associated with more specific recall and stronger within-participant reinstatement. Here, we tested whether shared ISC and ISPS predict recall specificity and within-participant reinstatement, and whether these relationships differ across younger and older adults. Younger and older adults watched a 26-minute episode of a "slice-of-life" television sitcom (HBO's Curb Your Enthusiasm) during fMRI scanning and subsequently provided an unconstrained spoken recall. We combined three complementary analyses centered on the PMN: (1) Within-participant event-specific pattern reinstatement between encoding and recall to assess age-related differences in event reactivation; (2) ISC during encoding to quantify age-related differences in temporal synchrony of PMN responses; and (3) ISPS computed for encoding-to-encoding, encoding-to-recall, and recall-to-recall event pairs to assess the degree to which event representations are shared across individuals. Critically, ISC and ISPS was evaluated relative to within- and cross-age group, allowing us to test which reference group provides the most informative basis for successful memory for both younger and older adults.

Prior work on younger adults has shown that closer alignment with age-matched individuals predicts better memory (Simony et al., 2016; Dmochowski et al., 2014), consistent with the view that age-typical processing scaffolds effective encoding and retrieval. For older adults, however, the optimal reference group for predicting successful memory is less clear. One possibility is that alignment with other older adults remains beneficial, reflecting age-appropriate event processing. Alternatively, given that older adults exhibit greater idiosyncrasy in neural

responses (Campbell et al., 2015; Geerligs et al., 2018), similarity to other older adults may be less diagnostic of successful processing. Instead, memory may be strongest in individuals whose response profiles more closely resemble patterns observed in younger adults, consistent with cognitive reserve accounts (Stern 2012; Cabeza et al., 2018). A third possibility is that successful memory reflects convergence on a shared event representation that generalizes across age groups best predicts reinstatement and recall specificity.

Our findings reveal that aging alters how naturalistic events are encoded, recalled, and reinstated during recall. Older adults recalled the episode with reduced specificity and showed weaker event-specific reinstatement compared to younger adults. Older adults also exhibited reduced moment-to-moment neural synchrony (ISC) and decreased overlap of event-level spatial activation patterns (ISPS) within their own age group during encoding, while cross-age group comparisons showed no differences. This coincides with prior work indicating that aging is associated with idiosyncratic event representations as experiences unfold (Campbell et al., 2015; Geerligs et al., 2018). We extend this line of research by demonstrating that encoding-to-encoding ISPS was the most consistent predictor of reinstatement across both age groups. Reinstatement in younger adults, however, was additionally predicted by ISC and encoding-to-recall ISPS, whereas in older adults, reinstatement was most strongly predicted by the degree to which encoding patterns resembled those of younger adults. Together, these results suggest that age-related reductions in neural reinstatement reflect altered encoding dynamics and diminished convergence on shared event representations, and that preserving encoding patterns similar to those of younger adults may support more detailed recall and more successful reinstatement among older adults.

RESULTS

Age-related Differences in Recall Specificity

Twenty-seven younger (M = 21.96, SD = 2.83; range = 18-28; 18 female; 12 Asian, 3 Black, 5 White, 7 Prefer Not to Answer/Not Listed) and twenty-three older (M = 68.43, SD = 6.82; range = 60-83; 11 female; 2 Asian, 18 White, 3 Prefer Not to Answer) adult participants were presented with an "slice-of-life" television sitcom (HBO's *Curb Your Enthusiasm*) while undergoing functional MRI scanning (Fig. 1A). After viewing, participants were instructed to recall aloud everything that they can remember about the television episode in as much detail as possible (Fig. 1B). Recall was recorded with an MRI-compatible microphone and later transcribed and manually scored for the specificity of recall against a standardized rubric.

The television episode was divided into high-level meaningful transitions (Zacks et al., 2007), resulting in 37 possible coarse-grained events and 195 details descriptions for each event (Delarazan et al., 2025). Two raters (E.K. and H.C.) independently segmented each participants' recall and labeled each segment according to the event it pertained to. Each event was annotated with a structured description specifying who was involved, where it occurred, and what happened. The raters scored each recalled event for specificity by awarding gist units when a participant conveyed the general meaning of an event description and detail units when the participant additionally recalled specific components (e.g., who, where, what). Detail scoring was nested within gist such that any detail unit necessarily implied the corresponding gist unit, whereas gist units could be awarded in the absence of detail. Recall specificity was calculated as:

$$Recall\ Specificity = 1 + \left(\frac{Detail\ Units - Gist\ Units}{Detail\ Units + Gist\ Units}\right)$$

where *Detail Units* and *Gist Units* denote the total number of detail and gist units, respectively. This ratio reflects the relative balance of detailed versus gist-based recall, with higher values

indicating more specific recall (see Supplementary Materials). Intraclass correlation coefficient (ICC) assessing the agreement between two raters revealed high-level of agreement (Single raters: ICC = 0.96, 95% CI [0.77, 1.0], $F(5,5) = 56.09$, $p < .001$; Average raters: ICC = 0.98, 95% CI [0.87,1.0], $F(5,5) = 56.09$, $p < .001$).

Our results revealed a significant difference in recall performance between younger (Min: 0.60, Max: 0.97, M = 0.75, SD = 0.19) and older (Min: 0.41, Max: 0.82, M = 0.63, SD = 0.11) adults ($t(42.76) = 2.73$, Cohen's d = 0.75, $p = .009$, BF10 = 5.28, 95% CI = [0.03, 0.21]; Fig. 1C), indicating that older adults tended to recall event details in a more gist-like manner compared to younger adults. This pattern converges with prior work showing age-related reductions in episodic specificity, with older adults preserving overall meaning while losing access to fine-grained contextual details (Schacter et al., 1991; Castel, 2005; Greene & Naveh-Benjamin, 2023; Melega et al., 2025; Delarazan et al., 2025; Tinner, 2026).

Higher Event-specific Neural Reinstatement Among Younger Adults

Recent evidence suggests that younger adults tend to engage a posterior medial network (PMN) of cortical areas when reconstructing and experiencing ongoing naturalistic experiences (Chen et al., 2017; Zabood et al., 2017). With this, all our analyses focused *a priori* on regions of the PMN, which include the posterior cingulate cortex, precuneus, and lateral parietal regions (Ritchey & Ranganath, 2012; Barnett et al., 2021). Prior work involving younger adults has shown that neural patterns of activity when one experiences an ongoing event within the PMN are reactivated during retrieval of that event. Here, we first examined whether event-specific neural reinstatement within an individual diminishes among older adults.

For each participant, neural timeseries data was aligned with timepoints (e.g., onset and offset) of events during encoding and recall and shifted +5 TRs (~6.1 s) to account for

hemodynamic lag in the BOLD signal. Timepoints within each event were averaged, resulting in one pattern of neural activity for each event. Neural patterns for each event during encoding were correlated with neural patterns for each event during spoken recall using Pearson's r and Fisher's z-transformation to stabilize variance and then averaged across vertices and hemispheres. To account for general neural similarity unrelated to event-specific reinstatement, we used the pattern similarity between mismatched encoding to recall event pairs (e.g., encoding of event A to recall of event B) as a baseline and subtracted this value from the pattern similarity values for matched encoding to recall event pairs (e.g., encoding of event A to recall of event A). Baseline-corrected event-specific neural reinstatement values were estimated within-participant, and higher values reflect greater similarity between encoding and recall neural patterns relative to baseline, suggesting stronger event-specific neural reinstatement (Fig. 2A).

Baseline-corrected reinstatement values were tested against zero separately for each age group using one-sample permutation tests with 5,000 iterations to confirm that encoding-recall pattern similarity was reliably greater than 0. Effect size was quantified using one-sample Cohen's d, and 95% confidence intervals were estimated using a bootstrap procedure including 10,000 iterations with replacement. We found that pattern reinstatement greater than zero in both younger ($M = 0.05$, $p < .001$, Cohen's $d = 1.27$, 95% CI = [0.03, 0.06]) and older ($M = 0.02$, $p < .001$, Cohen's $d = 0.83$, 95% CI = [0.01, 0.03]) adults.

Age group differences in event-specific pattern reinstatement were then assessed using a label-shuffling permutation test with 5,000 iterations. On each iteration, group labels were randomly shuffled and the mean difference in reinstatement was recomputed, generating a null distribution under the assumption that age group membership does not influence reinstatement. Effect size was quantified using Cohen's d, and 95% confidence intervals for the mean difference

were estimated using a bootstrap procedure including 10,000 iterations with replacement. Random seeds were fixed for reproducibility. Results revealed a significant age-related difference within the PMN such that younger adults' patterns of activity during recall were more similar to encoding of that matched event compared to older adults ($t = 2.29$, $p = .027$, Cohen's $d = 0.66$, 95% CI = [0.003, 0.040]; Fig. 2B). This reduced neural pattern reinstatement among older adults may reflect a weakened ability to reconstruct the specific structure of past experiences during recall.

Given that neural reinstatement depends on both the patterns during encoding and later recall, it remains unclear whether age-related reduction reflects age differences in encoding, recall, or both. One possibility is that reduced reinstatement within older adults reflects less shared event structure in PMN responses across individuals. Strong correspondence of neural responses across individuals is typically interpreted as evidence of a shared understanding of the core elements of events and their structure. On the other hand, weaker convergence suggests more idiosyncratic representations (Campbell et al., 2015; Geerligs et al., 2018; Cohen et al., 2018), which in turn may fail to ideally capture the meaning and structure of an event. Thus, idiosyncratic event-level patterns may be less likely to be reinstated within an individual. Therefore, we conducted a series of between-participant analyses to assess whether PMN responses were synchronized across individuals during encoding and during recall, and whether the degree of synchronicity differed with age.

Reduced Intersubject Correlation Among Older Adults During Encoding

One common approach to assess shared event structure during encoding is intersubject correlation (ISC) which measures how similarly individuals respond to the same time-locked stimulus (Hasson et al., 2004). ISC reflects the degree to which moment-by-moment fluctuations in neural activity are synchronized across individuals and is often taken as an index of shared

processing or interpretation of an unfolding ongoing experience (Hasson et al., 2004; Nastase et al., 2019). For each participant, preprocessed functional timeseries data were projected to PMN cortical surface space. We then correlated that participant's timeseries with the average timeseries of the remaining participants using a leave-one-out approach and computing Pearson's r. Resulting correlation coefficients were then Fisher z-transformed to stabilize variance and values were averaged across vertices and hemispheres. This procedure yielded one ISC value per participant within the PMN, where higher values reflect stronger temporal alignment and greater shared neural synchrony (Fig. 3A).

To ensure that ISC reflected a shared reliable signal rather than noise, we conducted circular time-shift permutation tests (Nastase et al., 2019) with 1000 iterations separately for each age group. Both older ($z = 0.092$, $p < .001$) and younger ($z = 0.146$, $p < .001$) adults showed ISC significantly greater than zero (Supplementary Fig. 1), confirming reliable shared neural responses during encoding. Group comparisons were then assessed using a label-shuffling permutation test with 5,000 iterations. Within-group ISC was significantly higher for younger than older adults ($t = -4.37$, $p < .001$, Cohen's $d = 1.21$, 95% CI = [0.030, 0.077]), while cross-group ISC did not differ between younger ($M = 0.108$, $SD = 0.036$) and older ($M = 0.101$, $SD = 0.044$) adults ($t = -0.66$, $p = .506$, Cohen's $d = 0.19$, 95% CI = [-0.015, 0.029]; Fig. 3B).

Consistent with prior work demonstrating reduced neural synchrony in older adults during naturalistic viewing (Campbell et al., 2015; Geerligs et al., 2018; Sternin et al., 2023), our results indicate that PMN responses are less reliably shared across older adults within their age group as they encode the same stimulus. This reduced synchrony suggests that older adults demonstrate more idiosyncratic PMN responses to the unfolding of ongoing experience, which may limit the formation of a stable event representation that can later be reinstated.

Age-related Changes in Intersubject Pattern Similarity

Together, our ISC and within-participant pattern reinstatement results indicate that aging reduces both event-specific reactivation in the PMN and temporal synchrony across individuals during encoding. Importantly, these measures do not reveal whether event representations are expressed in a common spatial pattern across individuals, which could reveal shared event comprehension and provide a stable basis for later reinstatement. We next investigated whether age-related differences also emerge in the shared representational structure of individual events by examining intersubject pattern similarity (ISPS). ISPS compares neural patterns for the same event across different participants, providing an index of how similar the spatial event representations are across brains (Chen et al., 2017).

Using the same event-averaged PMN patterns described above for within-participant neural pattern reinstatement, we correlated each participant's pattern for a given event with the average pattern of the remaining participants for the corresponding event using Pearson's r in a leave-one-out approach. The resulting correlation coefficients were Fisher z-transformed for group-level analyses and baseline-corrected by subtracting mismatched event pairs from matched event pairs. We computed ISPS within the PMN for encoding-to-encoding (Fig. 4A), encoding-to-recall (Fig. 4C), and recall-to-recall (Fig. 4E) event pairs, separately within each age group and across age groups. For each ISPS analysis, we first confirmed that event-specific ISPS was significantly greater than zero, then assessed age-related differences using a label-shuffling permutation test with 5,000 iterations comparing younger and older adults on within-group and cross-group ISPS separately. Higher values indicate that participants had more similar PMN patterns for the same event across individuals.

Encoding-to-encoding ISPS. To assess how consistently participants encoded the events from the television episode, we correlated each participant's event encoding patterns with group-averaged event encoding patterns from the remainder of participants in the same age group (within-age group) and participants in the other age group (cross-age group). Event-specific within-age group encoding-to-encoding ISPS was significantly greater than zero for both older ($M = 0.090$, $p < .001$, Cohen's $d = 2.91$, 95% CI = [0.078, 0.103]) and younger adults ($M = 0.167$, $p < 0.0002$, Cohen's $d = 2.78$, 95% CI = [0.143, 0.188]). We found similar results cross-age group encoding-to-encoding ISPS (Younger Adults: $M = 0.127$, $p < 0.001$, Cohen's $d = 2.88$, 95% CI = [0.111, 0.143]; Older Adults: $M = 0.113$, $p < 0.001$, Cohen's $d = 2.51$, 95% CI = [0.096, 0.132]). Together, these results confirm reliable shared encoding patterns in both groups and comparisons. Label-shuffling permutation tests comparing age groups revealed that younger adults exhibited substantially greater within-group encoding-to-encoding ISPS than older adults ($t = -5.74$, $p < .001$, Cohen's $d = 1.55$, 95% CI = [0.050, 0.100]; Fig. 4B). Cross-group encoding-to-encoding ISPS did not differ between younger and older adults ($t = -1.10$, $p = .290$, Cohen's $d = 0.31$, 95% CI = [-0.011, 0.037]).

Together, these results indicate that younger adults demonstrate similar encoding patterns to one another in the PMN, whereas older adults show relatively more idiosyncratic encoding patterns compared to others in their age group. Importantly, the absence of a group difference in cross-group ISPS mirrors our ISC findings and suggests that older adults do not encode events in a fundamentally different way from younger adults but instead show greater variability in their encoding patterns within their age group.

Encoding-to-recall ISPS. Next, to assess how consistently event representations were reinstated from encoding to recall across individuals, we correlated each participant's event-

specific recall patterns with group-averaged event-specific encoding patterns from the remainder of participants in the same age group (within-age group) and participants in the other age group (cross-age group). Within-group event-specific encoding-to-recall ISPS was significantly greater than zero for both older adults ($M = 0.037$, $p < 0.001$, Cohen's $d = 2.03$, 95% CI = [0.029, 0.044]) and younger adults ($M = 0.046$, $p < 0.001$, Cohen's $d = 1.61$, 95% CI = [0.035, 0.056]). Cross-group event-specific encoding-to-recall ISPS was also greater than zero among older and younger adults (Younger Adults: $M = 0.040$, $p < 0.001$, Cohen's $d = 1.97$, 95% CI = [0.033, 0.048]; Older Adults: $M = 0.039$, $p < 0.001$, Cohen's $d = 1.81$, 95% CI = [0.029, 0.048]). These results confirm reliable within- and cross-age group participant encoding-to-recall pattern similarity in both groups. Label-shuffling permutation tests revealed no significant age difference in within-age group ISPS ($t = -1.32$, $p = .20$, Cohen's $d = 0.37$, 95% CI = [-0.004, 0.022]) or cross-age group ISPS ($t = -0.29$, $p = .78$, Cohen's $d = 0.09$, 95% CI = [-0.010, 0.013]; Fig. 4D).

Together, these findings suggest that, in the PMN, younger and older adults show a broadly similar pattern of encoding-to-recall similarity across individuals. This stands in contrast to age-related reductions in encoding-to-encoding similarity patterns described earlier. In other words, although aging reduces within-participant reinstatement and neural synchrony during encoding, the way recall patterns relate to group-averaged encoding patterns appears largely preserved across age.

Recall-to-recall ISPS. Lastly, we assessed how similar neural patterns in the PMN during recall were among individuals by correlating each participant's event-specific recall patterns with group-averaged event-specific recall patterns from the remainder of participants in the same age group (within-age group) and participants in the other age group (cross-age group). Event-specific recall-to-recall ISPS was significantly greater than zero for both older adults ($M = 0.038$, $p < 0.001$,

Cohen's d = 1.81, 95% CI = [0.029, 0.047]) and younger adults (M = 0.049, p < 0.001, Cohen's d = 1.97, 95% CI = [0.039, 0.057]) when compared within their own age group. Cross-age group event-specific recall-to-recall ISPS was also significantly greater than zero in both age groups (Younger Adults: M = 0.030, p < 0.001, Cohen's d = 1.47, 95% CI = [0.022, 0.037]; Older Adults: M = 0.037, p < 0.001, Cohen's d = 1.63, 95% CI = [0.027, 0.047]), confirming reliable shared recall patterns. Label-shuffling permutation tests revealed no significant age difference in within-age group ISPS (t = -1.55, p = .13, Cohen's d = 0.45, 95% CI = [-0.003, 0.023]) or cross-age group ISPS (t = 1.12, p = .27, Cohen's d = -0.34, 95% CI = [-0.012, 0.005]; Fig. 4F). These findings suggest that the degree to which PMN recall patterns are shared across individuals is broadly similar for younger and older adults.

To summarize, our ISPS analyses revealed a dissociation between encoding and recall in how age affects shared neural representations in the PMN. Younger adults showed significantly greater within-age group encoding-to-encoding ISPS than older adults, indicating more consistent spatial patterns during initial encoding. However, this age difference did not extend to encoding-to-recall or recall-to-recall ISPS, where both groups showed comparable levels of shared pattern similarity. This pattern suggests that age-related differences in shared neural representations are primarily driven by variability during encoding rather than during recall. These results converge with our ISC findings where older adults show more idiosyncratic neural responses during encoding, but the way these patterns are expressed during recall is similarly structured across age groups. That is, our findings collectively suggest that encoding is the primary source of age-related divergence in shared PMN representations.

Relationship Between Shared Neural Processing and Pattern Reinstatement

Given that our ISPS analyses revealed age-related differences specifically during encoding, we next asked whether shared neural processing during encoding was associated with individual differences in pattern reinstatement. For each participant, we correlated their ISC and ISPS values (within- and cross-age group) with their baseline-corrected pattern reinstatement scores using Spearman's rho correlation, separately for each age group. Spearman's rho was chosen given the modest sample sizes, as it is robust to outliers and does not assume linearity. Statistical significance was corrected for multiple comparisons using false discovery rate (FDR) across all neural predictors ($q = 0.05$).

Across both age groups, encoding-related measures were the most consistent predictors of pattern reinstatement, while no recall-to-recall ISPS measures significantly predicted reinstatement in either group after correction (Table 1). Among younger adults, within-age group encoding-to-recall ISPS showed the strongest association with pattern reinstatement (Spearman's rho = .61, $p < .001$ uncorrected, $p = .004$ corrected), followed by within-age group encoding-to-encoding ISPS (Spearman's rho = .593, $p = .012$ uncorrected, $p = .004$ corrected) and cross-age group encoding-to-encoding ISPS (Spearman's rho = .609, $p = .002$ uncorrected, $p = .004$ corrected). ISC was also significantly associated with reinstatement for both within-age group (Spearman's rho = .432, $p = .024$ uncorrected, $p = .038$ corrected) and cross-age group (Spearman's rho = .436, $p = .024$ uncorrected, $p = .038$ corrected) comparisons for younger adults. Broadly, these findings align with the view that shared responses to an encoded stimulus relate to successful reinstatement of those neural patterns during retrieval. Among older adults, encoding-to-encoding ISPS was similarly predictive, with cross-age group (Spearman's rho = .672, $p = .002$ uncorrected, $p = .016$ corrected) and within-age group (Spearman's rho = .565, $p = .012$

uncorrected, p = .048 corrected) associations surviving correction. Unlike younger adults, ISC did not significantly predict reinstatement among older adults after correction.

Interestingly, a suggestive dissociation emerged for encoding-to-recall ISPS such that among younger adults, within-age group encoding-to-recall ISPS was a strongest predictor of pattern reinstatement, whereas cross-age group encoding-to-recall ISPS was not significant (Spearman's rho = .297, p = .133 uncorrected, p = .152 corrected). Among older adults, the cross-age group encoding-to-recall ISPS showed a moderate association to reinstatement (Spearman's rho = .498, p = .030 uncorrected, p = .080 corrected), while within-age group encoding-to-recall ISPS showed no relationship (Spearman's rho = .177, p = .468 uncorrected, p = .535 corrected). Although the older adult cross-age association did not survive FDR correction, the opposing pattern across age groups suggests that alignment with younger adults' encoding structure may be particularly relevant for reinstatement in older adulthood.

Together, these results indicate that participants whose encoding patterns more closely resembled those of other individuals showed stronger event-specific within-participant neural reinstatement. This relationship was driven primarily by encoding-related shared representations rather than recall-related similarity. This suggests that encoding plays an important role in individual differences in within-participant pattern reinstatement. Notably, among older adults, the strongest predictor of reinstatement was cross-age group encoding-to-encoding ISPS, suggesting that older adults whose encoding patterns more closely resembled those of younger adults showed the strongest neural reinstatement.

Relationship Between Shared Neural Processing and Recall Specificity

Having established that encoding-related shared neural measures predict neural reinstatement, we next asked whether these measures also predict the quality of behavioral recall.

For each participant, we computed a recall specificity ratio score reflecting the degree of detail versus gist units produced during spoken recall. We correlated all our neural measures (i.e., pattern reinstatement, ISC, and ISPS) with recall specificity using Spearman's rho correlation, and corrected for multiple comparisons using FDR (q = .05).

Of our measures, we observed a trending positive correlation between within-age group ISC and recall specificity among younger adults (Spearman's rho = 0.380, p = .051 uncorrected, p = .459 corrected) consistent with prior studies' suggestion that ISC reflects stimulus comprehension (Poulsen et al., 2017; Cohen et al., 2018; Liu et al., 2025). In contrast, no such relationship was observed in older adults (Spearman's rho = -0.035, p = .885 uncorrected, p = .960 corrected). These results suggest that for younger adults, common neural signals across individuals during encoding of details of the television episode supported more specific recall. The lack of such a relationship in older adults suggests that more idiosyncratic encoding signals may be detrimental to recall specificity. We caution, however, that the relationship between ISC and recall specificity in young adults does not survive FDR correction for multiple comparisons (q = 0.05, p = .459). No other neural measures significantly correlated with recall specificity among younger and older adults (Table 2).

Together, these findings reveal a dissociation between neural and behavioral outcomes. While shared encoding patterns robustly predict neural reinstatement in both age groups, they do not directly predict the specificity of verbal recall. This suggests that the neural processes captured by ISC and ISPS may support the formation of memory traces that can be reinstated. However, the specificity of what is ultimately expressed during recall likely depends on additional factors beyond shared neural encoding that are not well-captured by these measures.

DISCUSSION

Remembering everyday experiences often involves encoding of stable event representations and later reconstructing them during retrieval. Here, we examined how aging affects these processes by measuring neural activity in the posterior medial network (PMN) while younger and older adults watched and later freely recalled a television episode. Our findings show that aging alters both within-individual neural reinstatement and across-individual convergence of event representations. Older adults recalled the episode with reduced specificity and exhibited weaker event-specific reinstatement, alongside reduced temporal synchrony (ISC) and reduced overlap of event-level spatial patterns (ISPS) during encoding. Critically, encoding-to-encoding ISPS was the most consistent predictor of within-individual reinstatement across both age groups, though the specific pattern of predictors differed by age. In younger adults, reinstatement was predicted by multiple encoding-related measures including ISC, encoding-to-encoding ISPS, and encoding-to-recall ISPS. In contrast, reinstatement was most strongly predicted by the degree to which encoding patterns closely resembled younger adults. Together, these findings suggest that age-related memory decline reflects weaker reinstatement and greater variability in event representations across individuals, and that memory reinstatement in older adults may depend on the degree to which encoding patterns converge on those shared by younger adults.

A key contribution of the current study is that the neural factors predicting successful reinstatement differed between age groups. For younger adults, reinstatement predicted by a broad set of encoding-related measures, including within-age group and cross-age group encoding-to-encoding ISPS, within-age group encoding-to-recall ISPS, and both within-age group and cross-age group ISC. This pattern is consistent with the idea that convergence on a common, event representation supports the formation of stable event traces that can later be reinstated (Koch et

al., 2020; Rugg & Srokova, 2024). In contrast, older adults' reinstatement was predominantly predicted by encoding-to-encoding ISPS. The strongest association was observed for cross-age group encoding-to-encoding ISPS, indicating that older adults whose neural patterns during encoding more closely resembled those of younger adults or the broader patterns showed significantly stronger within-person reinstatement. Within-age group encoding-to-encoding ISPS was also predictive, but ISC and recall-related ISPS measures did not significantly predict reinstatement among older adults.

This pattern was further reinforced by a suggestive dissociation in encoding-to-recall ISPS. Among younger adults, reinstatement was strongly predicted by the similarity between their recall patterns and other younger adults' encoding patterns, but not by similarity to older adults' encoding. Among older adults, pattern reinstatement was associated with how closely their recall patterns matched younger adults' encoding, while similarity to other older adults' encoding showed no relationship. Although the older adult association did not survive multiple comparison correction, the opposing directionality across groups is consistent with the interpretation that younger adults' encoding patterns capture a more reliable representation of event structure. When older adults' recall aligns with that structure, rather than with the more variable encoding patterns of their own age group, within-participant reinstatement is stronger.

One possibility for this age-related asymmetry is that maintaining neural processing patterns similar to younger adults represents a form of cognitive reserve or neural resilience that protects against age-related memory decline (Stern 2019; Cabeza et al., 2018). Under this account, older adults who deviate less from encoding patterns characteristic of younger adults preserve more efficient event processing, enabling them to form stronger memory traces that can be more successfully reinstated during retrieval. This interpretation aligns with theories proposing that

cognitive reserve reflects the preservation of effective neural response profiles despite age-related change (Stern et al., 2020). Alternatively, this pattern may reflect properties of the older adults' representation itself. Given that older adults show reduced within-age group ISC and ISPS, alignment with other older adults may provide a weaker or less stable scaffold for memory. In this case, the younger adult pattern may function as a more accurate representation of event structure. Older adults who converge on this broader pattern may benefit not because they are processing "like younger adults" per se, but because they are accessing a more robust, generalizable event representation that is less affected by the increased variability characteristic of older adult neural responses.

The encoding-to-recall ISPS dissociation offers additional evidence of this. Reinstatement among younger adults was predicted by the similarity between their recall patterns and the encoding patterns of other younger adults, but not those of older adults. Older adults, on the other hand, demonstrated that reinstatement was positively correlated with how closely their recall aligned with younger adults' encoding, but not other older adults'. Although the association in older adults did not survive multiple comparison correction, the opposing directionality across groups is more consistent with the idea that it is may not simply be that older adults who process events similarly to younger adults show better reinstatement. Instead, it shows that reinstatement in both age groups tracks alignment with the same encoding signal which is one that is better captured by the younger adult group. This suggests that younger adults' encoding patterns may approximate a more accurate representation of the shared event structure, and that successful reinstatement depends on the degree to which recall converges on that structure regardless of age. Clarity on the nature of this age-related loss of neural synchrony, however, will require longitudinal and individual differences approaches. Nevertheless, our findings highlight that

successful memory in older adulthood may depend less on convergence with older adult group patterns and more on preserved access to generalizable event representations.

One plausible source of this age-related difference in predictors is increased variability in how events are processed during encoding. Our ISC analyses revealed that older adults showed reduced moment-to-moment synchronicity of PMN responses during naturalistic viewing compared to younger adults. Importantly, this age difference was specific to within-age group ISC, where younger adults showed significantly greater synchrony than older adults. Cross-age group ISC did not differ between groups, suggesting that older adults do not process the movie in a fundamentally different way from younger adults but instead show greater interindividual variability within their age group.

This reduced synchrony suggests that older adults process the unfolding event more idiosyncratically, potentially reflecting differences in attention allocation, event segmentation, or the construction of situation models (Zacks et al., 2006; Reagh et al., 2020). Importantly, this finding is aligned with prior work demonstrating age-related reductions in neural synchrony during movie viewing (Campbell 2015; Greerligs et al., 2018; Sternin et al., 2021). We note that such differences are particularly pronounced in the PMN, a network of regions critical for representing event structure and integrating information across time (Ranganath & Ritchey, 2012; Ritchey & Cooper, 2020; Reagh & Ranganath 2023). The reduced synchrony we observed may reflect a constraint on memory formation, such that less coherent event structure during encoding yields weaker, less detailed representations that are less accessible, underspecified, or are otherwise harder to reinstate during recall.

Our ISPS analyses revealed age-related changes in how patterns of neural activity in the PMN represent events. During encoding, younger adults showed substantially higher within-age

group pattern similarity than older adults, indicating that younger adults form more consistent spatial patterns of neural activity in response to the same events. As with ISC, cross-age group encoding-to-encoding ISPS did not differ between groups, further confirming that the age difference is driven by within-age group variability rather than fundamentally different encoding strategies. In contrast, older adults showed greater variability in how events are represented across individuals within their age group.

This finding aligns with neural dedifferentiation accounts, which proposes that aging leads to less distinctive, dedifferentiated, or generally more variable neural representations (Koen & Rugg, 2019; Park et al., 2004; Carp et al., 2011; St-Laurent et al., 2014). Our results extend this framework by demonstrating that dedifferentiation manifests not only as reduced distinctiveness within individuals, but also as reduced consistency across individuals viewing the same events. This encoding-centered account converges with evidence that age differences in cortical reinstatement are largely driven by reduced neural differentiation at encoding rather than retrieval-specific deficits (Hill et al., 2021; Pauley et al., 2023). Increased interindividual variability suggests that aging weakens the constraints that normally guide neural responses toward a common representational format when processing shared experiences. Critically, these age differences in shared representations were most pronounced during encoding and when comparing individuals within their own age group. In contrast, ISPS for encoding-to-recall and recall-to-recall event pairs showed comparable magnitudes across age groups, suggesting that the primary deficit lies in the initial formation of event representations rather than in how strongly recall patterns align with shared event structure. Taken together, our encoding-centered results suggest that when event structure is established less consistently across individuals, the downstream representations

available for later reinstatement may be weaker which could help explain why older adults show weaker event-specific reinstatement during recall.

Our pattern of results, however, indicates that the story is more nuanced than simple encoding failure. The preservation of encoding-to-recall and recall-to-recall ISPS at comparable levels across age groups suggests that older adults can still express recall patterns that align with the group-level event structure. In other words, even though older adults showed reduced within-participant reinstatement, their recall patterns still resembled the spatial structure observed at encoding across individuals. The dissociation between preserved group-level ISPS and reduced within-participant reinstatement suggests that aging impacts the fidelity of event representations more than the capacity to express event-related patterns during recall. These findings build on prior work demonstrating age-related reductions in reinstatement (Hill et al., 2021; Trelle et al., 2020) by revealing that the deficit reflect both less consistent representations during encoding and weaker reactivation of those representations during recall.

Moreover, this dissociation also helps reconcile the neural findings with the behavioral shift toward gist. Preserved group-level similarity in event representations does not necessarily imply that memories are retrieved with high specificity. Instead, our results suggest that older adults may be able to broadly reconstruct events sufficient for recall patterns to align with shared structure while losing the fine-grained contextual details that support highly specific remembering. Consistent with this account, we found a trend-level association between neural synchrony during encoding predicted recall specificity in younger adults but not older adults. Among younger adults, higher within-age group ISC was associated with more detailed, specific recall, suggesting that individuals whose neural responses more closely tracked the unfolding narrative were better able to encode and later recover event details (Hasson et al., 2008; Nguyen et al., 2019). While this

association did not reach significance and did not survive correction for multiple comparisons, and thus should be interpreted cautiously, it aligns with prior work linking neural synchrony during naturalistic viewing to memory success in younger adults.

The absence of this relationship in older adults is notable. One possibility is that, even when older adults' neural responses are temporally aligned with others, this synchrony does not reflect the same degree of detailed event encoding observed in younger adults. Alternatively, the processes that give rise to synchrony may differ across age such that ISC may reflect moment-by-moment construction of a detailed situation model among younger adults, whereas in older adults it may reflect more general engagement with the stimulus without comparable precision in representing contextual features. In both cases, this interpretation is consistent with our behavioral finding that older adults recalled the episode less specifically, relying more heavily on general thematic information than on precise contextual details. This pattern aligns with extensive evidence that aging shifts memory toward gist (Schacter et al., 2012; Kouststaal & Schacter, 1997; Grilli & Sheldon, 2022). Older adults often retain the meaning and structure of experiences while losing fine-grained perceptual and temporal detail (St-Laurent et al., 2011; Addis et al., 2008). Although this shift can be adaptive in some contexts (Hess 2005), our findings suggest it may also reflect changes in how events are encoded and stabilized in the first place. That is, when event representations are less coherent or less consistent during encoding later retrieval may disproportionately draw on the more robust, gist-like components that were successfully captured. Thus, variability at encoding may reduce representational detail and shift recall toward gist.

Finally, while prior work has focused on age-related changes in the hippocampus and medial temporal lobe (Raz et al., 2005; Fjell et al., 2014), our results highlight that functional alterations in large-scale cortical networks play an equally important role. We focused a priori on

the PMN due to its involvement in constructing situation models and integrating information across extended timescales. Our findings suggest that age-related memory change may partly reflect reduced stability in PMN event representations as experiences unfold. Importantly, the PMN is also implicated in vulnerability to neurodegenerative disorders. While our sample was not designed to test disease-related hypotheses, converging work suggests that regions overlapping the PMN are among the earliest sites of amyloid-β accumulation (Palmqvist et al., 2017; Sperling et al., 2009; Buckner et al., 2008) and show early disruption in Alzheimer's disease-related pathways (Braak & Braak, 1991; Maass et al., 2017; Harrison et al., 2019; Therriault et al., 2022). Our results motivate future work that can combine naturalistic memory paradigms such as this with targeted measures of age-related biomarkers. These studies could test whether PMN disruption and age-related differences in shared encoding patterns track preclinical Alzheimer's disease risk and help explain variability in memory outcomes among older adults.

Together, our study shows that event memories reflecting real-world experiences depend on building coherent event representations as experiences unfold and reactivating those representations during recall. Aging was associated with weaker event-specific reinstatement in the PMN and reduced convergence on shared event representations during encoding, alongside a shift toward less specific recall. Interestingly, the shared neural structure that supported reinstatement differed by age. In younger adults, stronger reinstatement was associated with multiple encoding-related measures of shared processing, whereas in older adults, reinstatement was strongest when encoding patterns resembled those of younger adults. These results suggest rather than reflecting retrieval alone, reinstatement may be bounded by encoding consistency and the degree to which a shared event structure is established during ongoing experience. Importantly,

our findings shift the emphasis in age-related memory decline toward the consistency of shared neural encoding patterns across individuals, informing future work on aging and cognitive decline.

METHODS

Transparency and Openness

Materials, primary data, and analysis scripts are publicly available on OSF Framework and Github. No aspect of the experiment was preregistered.

Participants

Participants were recruited from the University of California, Davis and the Davis, California community through online advertisement, flyers, and word of mouth. Participants provided written informed consent in accordance with the Institutional Review Board of the University of California, Davis, Project Title: Naturalistic Memory and Neurocognitive Aging (ID #: 1198619) protocol, and were compensated $20 per hour. Exclusion criteria included magnetic bodily implants, claustrophobia, a history of major psychiatric or neurological disorders, a concussion in the past 6 months, untreated diabetes or hypertension, current drug or alcohol abuse, uncorrected hearing, uncorrected vision, and age outside a range of 18-35 years for younger adults and age above 60 years for older adults.

Sixty-two participants were initially recruited to participate in the study: 39 younger adults and 26 older adults but excluded three younger adults and two older adults due to excessive motion in the MRI scanner, four younger adults due to unusable functional scans, two younger adults due to file transfer issues of scans, and one younger adult due to falling asleep during MRI scanning. Lastly, we excluded two younger adults and one older adult due to featuring greater than 80% of their whole-brain networks showing strong negative encoding-recall pattern similarity effects. This is a liberal threshold adopted to exclude participants showing extreme different event greater than same event pattern similarity effects. Briefly, we reasoned that such participants' data would be difficult to coherently interpret, given that noise more likely contributes to such a widespread

event differentiation effect than any underlying cognitively-relevant explanation. The final sample resulted in twenty-seven younger adults (M = 21.96, SD = 2.83; range = 18-28; 18 female; 12 Asian, 3 Black, 5 White, 7 Prefer Not to Answer/Not Listed) and twenty-three older (M = 68.43, SD = 6.82; range = 60-83; 11 female; 2 Asian, 18 White, 3 Prefer Not to Answer). A total of four older adult participants were excluded in analyses involving recall with three excluded due to corrupt audio files and one excluded excessive motion during the recall scan.

Stimuli

Encoding. Older and younger participants watched a twenty-six-minute episode of a television "slice-of-life" sitcom (HBO's Curb Your Enthusiasm, S01E07 AAMCO; Fig. 1A). The episode consisted of 37 events identified based on meaningful transitions (Zacks et al., 2007).

Free Recall. Participants were presented with a fixation cross in the center of a computer screen and were instructed to verbally recall everything that they can remember about the episode (Fig. 1B). Participants were encouraged to continue recalling for as long as they remembered additional details.

Neuropsychological Tests. Older participants also completed a battery of neuropsychological tests to test for cognitive impairments that might be relevant for assessing task performance (Table 3). These tests included: Craft21 Recall Immediate, Craft21 Recall Delayed, Montreal Cognitive Assessment (MoCA), and Multilingual Naming Test (MINT). Briefly, Craft21 assesses recall for narratives (Weintraub et al., 2018), MoCA coarsely assesses cognitive ability (Nasreddine et al., 2005), and MINT assesses the ability to name objects in English (Gollan et al., 2012).

Procedure

Participants were briefed on the experiment and provided written informed consent. Participants then completed an MRI screening form and demographics questionnaires. Upon successful screening for the experiment, participants were escorted to the Imaging Research Center and were prepared to undergo scanning.

Scanning took place in a Siemens 3 T Skyra with a 32-channel head coil. Participants were provided MRI-compatible earbuds with replaceable foam inserts (MRIaudio) as well as additional foam padding inside the head coil for hearing protection and to mitigate head motion. Participants were also given bodily padding, blankets, and corrective lenses as needed. An MRI-compatible microphone (Optoacoustics FOMRI-III) with bidirectional noise-canceling was affixed to the head coil, and the receiver (covered by a disposable sanitary pop screen) was positioned over the participant's mouth. Participants were given a description of strategies to remain still while speaking during functional image acquisition. During MRI data acquisition, an eye tracker was operational to monitor participants' wakefulness and head motion during spoken recall, but no eye tracking data were recorded.

High-resolution structural images and functional imaging runs were acquired (see *fMRI Data Acquisition*): One run of encoding the television episode, one run of free recall, and three runs of resting state scans (pre-encoding, post-encoding, and post-recall). Resting state scans were not analyzed in the current manuscript but will be included in a forthcoming paper. Briefly, participants were presented a fixation cross and instructed to relax and to freely think about anything for eight minutes during the resting state scans. After the scan, older adults completed a series of neuropsychological tests (Table 3).

fMRI Data Acquisition. High-resolution T1-weighted structural images were acquired using a magnetization prepared rapid acquisition gradient echo (MPRAGE) pulse sequence (FOV = 256mm, image matrix = 256 x 256, sagittal slices = 208, thickness = 1 mm). Functional images were acquired using a multi-band gradient echo planar imaging (EPI) sequence (TR = 1220 ms, TE = 24 ms, FOV = 192 mm, image matrix = 64 × 64, flip angle = 67, multi-band factor = 2, axial slices = 38, voxel size = 3 mm isotropic, P » A phase encoding, ACPC alignment). A single four TR functional scan of reverse phase encoding polarity (A » P) was acquired for unwarping (see *fMRI Data Preprocessing* below).

fMRI Data Preprocessing

Results included in this manuscript come from preprocessing performed using *fMRIPrep* 20.2.7 (RRID:SCR_016216; Esteban, Markiewicz, et al., 2018; Esteban, Blair, et al. 2018), which is based on *Nipype* v1.7.0 (RRID:SCR_002502; Gorgolewski et al., 2011). The T1-weighted (T1w) image was corrected for intensity non-uniformity (INU) with N4BiasFieldCorrection (Tustison et al., 2010), distributed with *ANTs* v2.3.3 (Avants et al., 2008, RRID:SCR_004757), and used as T1w-reference throughout the workflow. The T1w-reference was then skull-stripped with a *Nipype* implementation of the antsBrainExtraction.sh workflow from *ANTs* (using OASIS30ANTs). Brain tissue segmentation of cerebrospinal fluid (CSF), white-matter (WM) and gray-matter (GM) was performed on the brain-extracted T1w using fast from *FSL* v5.0.9 (RRID:SCR_002823; Zhang, Brady, & Smith, 2001). Brain surfaces were reconstructed using recon-all (FreeSurfer 6.0.1, RRID:SCR_001847, Dale, Fischl, & Sereno, 1999), and the brain mask estimated previously was refined with a custom variation of the method to reconcile ANTs-derived and FreeSurfer-derived segmentations of the cortical gray-matter of Mindboggle (RRID:SCR_002438; Klein et al., 2017).

A reference volume and its skull-stripped version were generated using a custom methodology of fMRIPrep. A B0-nonuniformity map (or fieldmap) was estimated based on two echo-planar imaging (EPI) references with opposing phase-encoding directions, using topup from *FSL*. Based on the estimated susceptibility distortion, a corrected EPI reference was calculated for a more accurate co-registration with the anatomical reference. The BOLD reference was then co-registered to the T1w reference using bbregister from *FreeSurfer*, which implements boundary-based registration (Greve & Fischl, 2009). Co-registration was configured with six degrees of freedom. Head-motion parameters with respect to the BOLD reference (transformation matrices, and six corresponding rotation and translation parameters) were estimated before any spatiotemporal filtering using mcflirt from *FSL* and functional runs were slice-time corrected using 3dTshift from *AFNI*. Functional timeseries were resampled onto the fsaverage surface using mri_vol2surf from *FreeSurfer*, linearly detrended, and denoised by regressing out the six motion parameters. For each participant, surface-based functional time-series were z-scored within vertex to normalize signal amplitude and reduce inter-subject variability.

Region of Interest (ROI)

Normalized surface-based time-series were parcellated using the Glasser HCP-MMP atlas (Glasser et al., 2016). Parcel-level network identity was then assigned using the cortico-hippocampal network classification derived from data-driven resting-state functional connectivity analyses (Barnett et al., 2021). From these subnetworks, our a priori region of interest was the Posterior Medial Network (PMN), which includes the posterior cingulate cortex, anterior retrosplenial cortex, posterior angular gyrus, and dorsal prefrontal cortex.

FIGURES

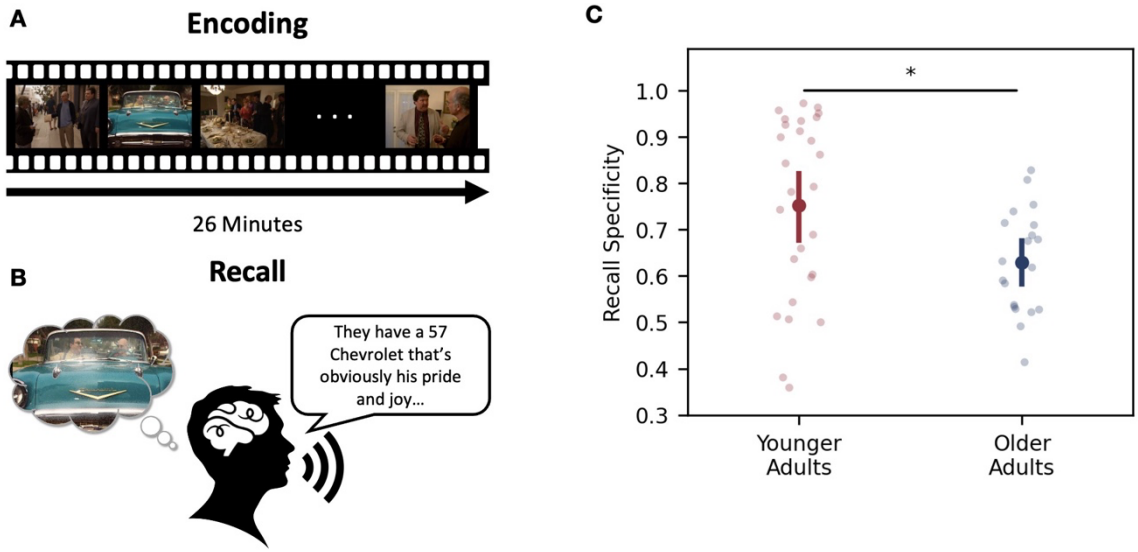

Figure 1. (A) Encoding. Participants were presented with an episode of a television sitcom (HBO's Curb Your Enthusiasm, S01E07: AAMCO). The episode consisted of 37 events. **(B) Recall.** Participants were then instructed to recall the episode in as much detail as possible. **(C) Recall Specificity.** Reflects the quality of recall with lower values indicating less detailed recall. **Key:** Solid dot represents group average performance. Error bars represent 95% confidence intervals. Points represent individual participants' average performance. Significant tests: n.s. > .05, * $p < .05$, ** $p < .01$, *** $p < .001$

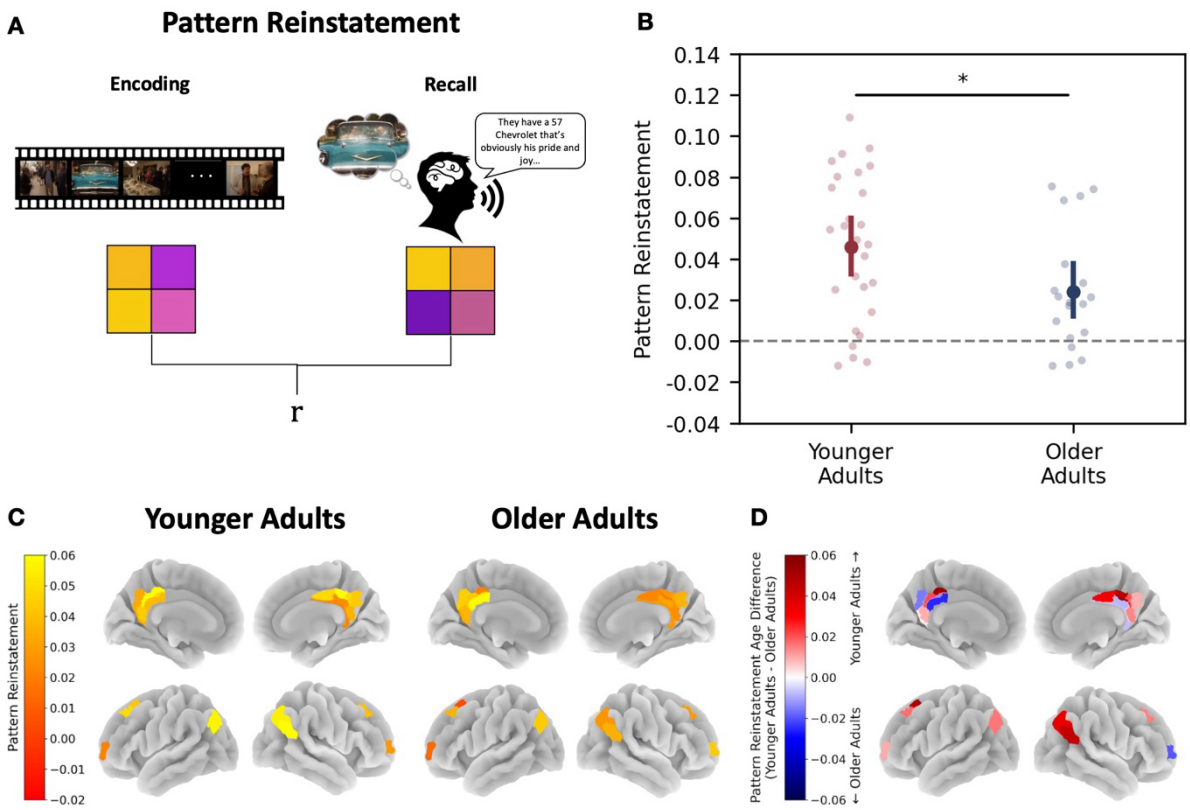

Figure 2. Pattern Reinstatement (A) Overview of Event-specific Neural Pattern Reinstatement. Schematic illustrating how neural activity was extracted for each event during encoding and free recall, averaged within event, and correlated using Pearson's r to quantify reinstatement strength. Resulting values were Fisher z–transformed and baseline-corrected by subtracting mismatched from matched encoding-to-recall event correlations, yielding an index of event-specific reinstatement. **(B) Event Specific Neural Pattern Reinstatement.** Pointplot comparing event-specific pattern reinstatement between younger and older adults. Higher values reflect greater similarity between encoding and recall activity patterns. **(C) Brain Map of Event-specific Neural Pattern Reinstatement**. Shows average encoding-to-recall event-specific reinstatement within the PMN for younger and older adults. Lighter colors values indicate stronger reinstatement, reflecting greater similarity between encoding and recall activity patterns. **(D) Difference Map of Event-specific Neural Pattern Reinstatement**. Displays the mean difference in reinstatement between younger and older adults. Warmer colors indicate stronger reinstatement in younger relative to older adults. **Key:** Solid dot represents group average performance. Error bars represent 95% confidence intervals. Points represent individual participants' average performance. Solid lines represent line of best fit and confidence interval. **Significant tests:** n.s. > .05, * p < .05, ** p < .01, *** p < .001

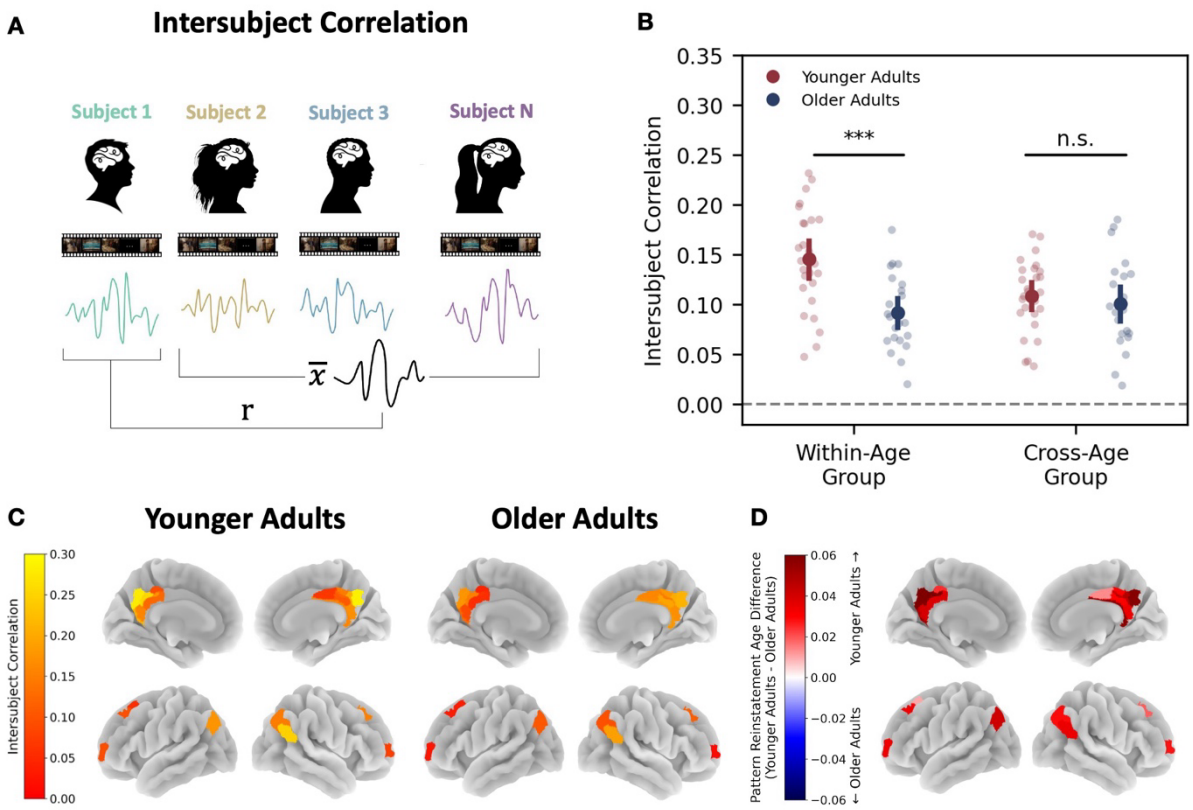

Figure 3. Intersubject Correlation (ISC). (A) Overview of ISC. Schematic illustrating how PMN timeseries for each participant were correlated with the average timeseries of all other participants within the same age group during encoding. Fisher z-transformed ISC values reflect the degree of moment-to-moment neural synchrony across viewers during movie watching. **(B) ISC.** Pointplot comparing intersubject correlation (ISC) during encoding for younger and older adults. Higher values indicate stronger temporal synchrony within age group. **(C) Brain Map of Within-Age Group ISC.** Shows average ISC values within the PMN for younger and older adults during encoding. Lighter colors values indicate stronger reinstatement, reflecting greater similarity between encoding and recall activity patterns. **(D) Difference Map of Within-Age Group ISC.** Shows the average difference in ISC between age groups. Warmer colors indicate greater temporal synchrony in younger relative to older adults. **Key:** Solid dot represents group average performance. Error bars represent 95% confidence intervals. Points represent individual participants' average performance. Solid lines represent line of best fit and confidence interval**. Significant tests:** n.s. > .05, * p < .05, ** p < .01, *** p < .001

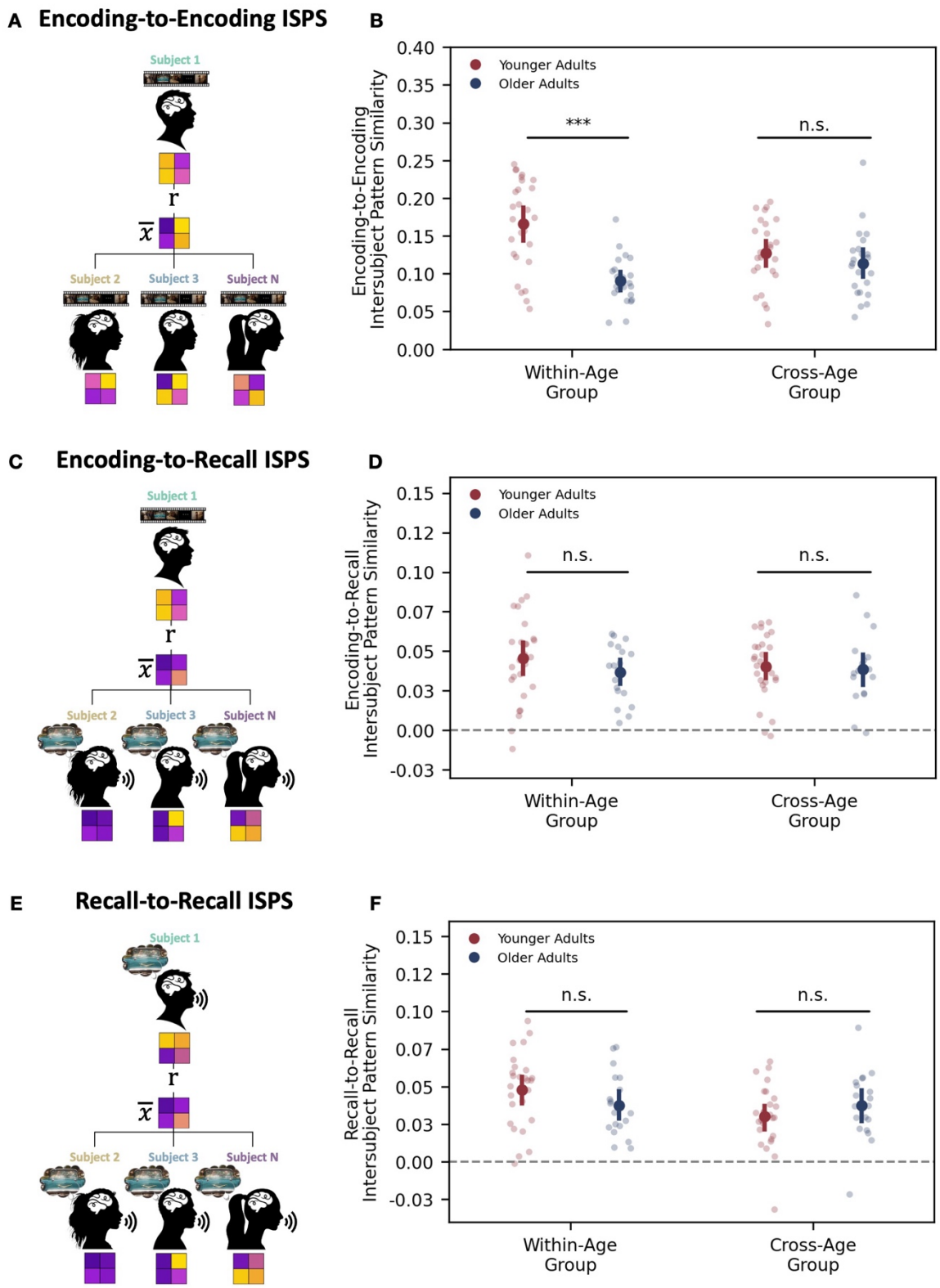

Figure 4. (A) Overview of Encoding-to-Encoding Intersubject Pattern Similarity (ISPS). Schematic illustrating how PMN event patterns during encoding were correlated across participants to assess the degree to which event representations were shared across brains. **(B) Encoding-to-Encoding ISPS.** Pointplots comparing encoding-to-

encoding ISPS within-age, across-age, and all-participant reference groups for younger and older adults. Higher values reflect more similar encoding representations across individuals. **(C) Overview of Encoding-to-Recall ISPS.** Schematic illustrating how encoding patterns were correlated with group-averaged recall patterns to assess the extent to which event representations formed during encoding generalize to retrieval across brains**. (D) Encoding-to-Recall ISPS.** Pointplots comparing encoding-to-recall ISPS within-age, across-age, and all-participant groupings for younger and older adults. Higher values reflect more similar shared representation from encoding to recall. **(E) Overview of Recall-to-Recall ISPS.** Schematic illustrating how recall patterns were correlated across participants to assess how similarly events were reconstructed at retrieval. **(F) Recall-to-Recall ISPS.** Pointplots comparing recall-to-recall ISPS within-age, across-age, and all-participant groupings for younger and older adults. Higher values reflect more similar recall representations across individuals. **Key:** Solid dot represents group average performance. Error bars represent 95% confidence intervals. Points represent individual participants' average performance. Solid lines represent line of best fit and confidence interval. **Significant tests:** n.s. > .05, * $p < .05$, ** $p < .01$, *** $p < .001$

TABLES

Table 1. Correlations between neural measures of intersubject correlation (ISC), and intersubject pattern similarity (ISPS) predicting pattern reinstatement.

Analysis	Younger Adults			Older Adults		
	rho	*p*	p_{FDR}	*rho*	*p*	p_{FDR}
Encoding-to-Encoding ISC Within-Age Group	0.432	.024*	.038*	0.407	.084	.112
Encoding-to-Encoding ISC Cross-Age Group	0.436	.023*	.038*	0.407	.084	.112
Encoding-to-Encoding ISPS Within-Age Group	0.593	.001**	.004**	0.565	.012*	.048*
Encoding-to-Encoding ISPS Cross-Age Group	0.556	.003**	.008**	0.672	.002**	.016*
Encoding-to-Recall ISPS Within-Age Group	0.609	.001**	.004**	0.177	.468	.535
Encoding-to-Recall ISPS Cross-Age Group	0.297	.133	.152	0.498	.030*	.080
Recall-to-Recall ISPS Within-Age Group	0.400	.038*	.051	0.432	.065	.112
Recall-to-Recall ISPS Cross-Age Group	0.193	.335	.335	0.135	.581	.581

Table 2. Correlations between neural measures of pattern reinstatement, intersubject correlation (ISC), and intersubject pattern similarity predicting specificity ratio.

Analysis	Younger Adults			Older Adults		
	rho	*p*	p_{FDR}	*rho*	*p*	p_{FDR}
Pattern Reinstatement	-0.060	.764	.764	0.012	.960	.960
Encoding-to-Encoding ISC Within-Age Group	0.380	.051^{+}	.459	-0.035	.885	.960
Encoding-to-Encoding ISC Cross-Age Group	0.306	.120	.525	-0.111	.640	.960
Encoding-to-Encoding ISPS Within-Age Group	0.211	.292	.525	-0.018	.940	.960
Encoding-to-Encoding ISPS Cross-Age Group	0.225	.260	.525	-0.191	.420	.960
Encoding-to-Recall ISPS Within-Age Group	0.14	.485	.573	-0.104	.673	.960
Encoding-to-Recall ISPS Cross-Age Group	0.221	.268	.525	-0.128	.601	.960
Recall-to-Recall ISPS Within-Age Group	-0.132	.510	.573	-0.137	.576	.960
Recall-to-Recall ISPS Cross-Age Group	-0.164	.415	.573	0.202	.408	.960

1 **Table 3.** Neuropsychological test scores from older adults.

Neuropsychological Tests				
	M	***SD***	***Range***	**Total Possible**
Montreal Cognitive Assessment (MoCA)	27.55	2.13	(23, 30)	30
Craft 21 Recall Immediate (Verbatim)	24.73	5.84	(13, 36)	44
Craft 21 Recall Immediate (Paraphrased)	18.45	3.39	(10, 24)	25
Craft 21 Recall Delayed (Verbatim)	21.77	7.00	(7, 36)	44
Craft 21 Recall Delayed (Paraphrased)	17.41	3.79	(8, 24)	25
Multilingual Naming Test (MINT)	31.27	1.32	(28, 32)	32

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REFERENCES

Abdulrahman, H., & Henson, R. N. (2016). Effect of trial-to-trial variability on optimal event-related fMRI design: Implications for Beta-series correlation and multi-voxel pattern analysis. NeuroImage, 125, 756-766.

Addis, D. R., Wong, A. T., & Schacter, D. L. (2008). Age-related changes in the episodic simulation of future events. Psychological science, 19(1), 33-41.

Avants, B. B., Epstein, C. L., Grossman, M., & Gee, J. C. (2008). Symmetric diffeomorphic image registration with cross-correlation: evaluating automated labeling of elderly and neurodegenerative brain. Medical image analysis, 12(1), 26-41.

Baldassano, C., Chen, J., Zadbood, A., Pillow, J. W., Hasson, U., & Norman, K. A. (2017). Discovering event structure in continuous narrative perception and memory. Neuron, 95(3), 709-721.

Barnett, A. J., Reilly, W., Dimsdale-Zucker, H. R., Mizrak, E., Reagh, Z., & Ranganath, C. (2021). Intrinsic connectivity reveals functionally distinct cortico-hippocampal networks in the human brain. PLoS biology, 19(6), e3001275.

Benoit, R. G., & Schacter, D. L. (2015). Specifying the core network supporting episodic simulation and episodic memory by activation likelihood estimation. Neuropsychologia, 75, 450-457.

Braak, H., & Braak, E. (1991). Neuropathological stageing of Alzheimer-related changes. Acta neuropathologica, 82(4), 239-259.

Buckner, R. L., Andrews-Hanna, J. R., & Schacter, D. L. (2008). The brain's default network: anatomy, function, and relevance to disease. Annals of the New York Academy of Sciences, 1124(1), 1-38.

Buckner, R. L., Sepulcre, J., Talukdar, T., Krienen, F. M., Liu, H., Hedden, T., ... & Johnson, K. A. (2009). Cortical hubs revealed by intrinsic functional connectivity: mapping, assessment of stability, and relation to Alzheimer's disease. Journal of neuroscience, 29(6), 1860-1873.

Cabeza, R., Albert, M., Belleville, S., Craik, F. I., Duarte, A., Grady, C. L., ... & Rajah, M. N. (2018). Maintenance, reserve and compensation: the cognitive neuroscience of healthy ageing. Nature Reviews Neuroscience, 19(11), 701-710.

Campbell, K. L., Grigg, O., Saverino, C., Churchill, N., & Grady, C. L. (2013). Age differences in the intrinsic functional connectivity of default network subsystems. Frontiers in aging neuroscience, 5, 73.

Campbell, K. L., Shafto, M. A., Wright, P., Tsvetanov, K. A., Geerligs, L., Cusack, R., ... & Villis, L. (2015). Idiosyncratic responding during movie-watching predicted by age differences in attentional control. Neurobiology of aging, 36(11), 3045-3055.

Carp J, et al. (2011) Age differences in neural distinctiveness revealed by multi-voxel pattern analysis. Neuroimage 56(2):736-743.

Chen, J., Leong, Y. C., Honey, C. J., Yong, C. H., Norman, K. A., & Hasson, U. (2017). Shared memories reveal shared structure in neural activity across individuals. Nature neuroscience, 20(1), 115-125.

Craik, F. I. (2020). Remembering: An activity of mind and brain. Annual review of psychology, 71(1), 1-24.

Craik, F. I., & Salthouse, T. A. (Eds.). (2011). The handbook of aging and cognition. Psychology press.

Dale, A. M., Fischl, B., & Sereno, M. I. (1999). Cortical surface-based analysis: I. Segmentation and surface reconstruction. Neuroimage, 9(2), 179-194.

Danker, J. F., & Anderson, J. R. (2010). The ghosts of brain states past: remembering reactivates the brain regions engaged during encoding. Psychological bulletin, 136(1), 87.

Delarazan, A. I., March, K., Markantonakis, E., Dy, J., & Reagh, Z. M. (in press). Aging shifts recall of naturalistic events from temporal to topic organization 10. Psychology & Aging.

Dmochowski, J. P., Bezdek, M. A., Abelson, B. P., Johnson, J. S., Schumacher, E. H., & Parra, L. C. (2014). Audience preferences are predicted by temporal reliability of neural processing. Nature communications, 5(1), 4567.

Esteban, O., Blair, R. W., Nielson, D. M., Varada, J. C., Marrett, S., Thomas, A. G., ... & Gorgolewski, K. J. (2019). Crowdsourced MRI quality metrics and expert quality annotations for training of humans and machines. Scientific data, 6(1), 30.

Esteban, O., Markiewicz, C. J., Blair, R. W., Moodie, C. A., Isik, A. I., Erramuzpe, A., ... & Gorgolewski, K. J. (2019). fMRIPrep: a robust preprocessing pipeline for functional MRI. Nature methods, 16(1), 111-116.

Eichenbaum, H. (2017). Prefrontal–hippocampal interactions in episodic memory. Nature Reviews Neuroscience, 18(9), 547-558.

Fjell, A. M., McEvoy, L., Holland, D., Dale, A. M., Walhovd, K. B., & Alzheimer's Disease Neuroimaging Initiative. (2014). What is normal in normal aging? Effects of aging, amyloid and Alzheimer's disease on the cerebral cortex and the hippocampus. Progress in neurobiology, 117, 20-40.

Geerligs, L., & Campbell, K. L. (2018). Age-related differences in information processing during movie watching. Neurobiology of Aging, 72, 106-120.

Glasser, M. F., Coalson, T. S., Robinson, E. C., Hacker, C. D., Harwell, J., Yacoub, E., ... & Van Essen, D. C. (2016). A multi-modal parcellation of human cerebral cortex. Nature, 536(7615), 171-178.

Gollan, T. H., Weissberger, G. H., Runnqvist, E., Montoya, R. I., & Cera, C. M. (2012). Self-ratings of spoken language dominance: A Multilingual Naming Test (MINT) and preliminary norms for young and aging Spanish–English bilinguals. Bilingualism: language and cognition, 15(3), 594-615.

Gordon, A. M., Rissman, J., Kiani, R., & Wagner, A. D. (2014). Cortical reinstatement mediates the relationship between content-specific encoding activity and subsequent recollection decisions. Cerebral Cortex, 24(12), 3350-3364.

Gorgolewski, K., Burns, C. D., Madison, C., Clark, D., Halchenko, Y. O., Waskom, M. L., & Ghosh, S. S. (2011). Nipype: a flexible, lightweight and extensible neuroimaging data processing framework in python. Frontiers in neuroinformatics, 5, 13.

Greve, D. N., & Fischl, B. (2009). Accurate and robust brain image alignment using boundary-based registration. Neuroimage, 48(1), 63-72.

Harrison, T. M., La Joie, R., Maass, A., Baker, S. L., Swinnerton, K., Fenton, L., ... & Jagust, W. J. (2019). Longitudinal tau accumulation and atrophy in aging and alzheimer disease. Annals of neurology, 85(2), 229-240.

Hasson, U., Chen, J., & Honey, C. J. (2015). Hierarchical process memory: memory as an integral component of information processing. Trends in cognitive sciences, 19(6), 304-313.

Hasson, U., Furman, O., Clark, D., Dudai, Y., & Davachi, L. (2008). Enhanced intersubject correlations during movie viewing correlate with successful episodic encoding. Neuron, 57(3), 452-462.

Hasson, U., Landesman, O., Knappmeyer, B., Vallines, I., Rubin, N., & Heeger, D. J. (2008). Neurocinematics: The neuroscience of film. Projections, 2(1), 1-26.

Hasson, U., Nir, Y., Levy, I., Fuhrmann, G., & Malach, R. (2004). Intersubject synchronization of cortical activity during natural vision. science, 303(5664), 1634-1640.

Hess, T. M. (2005). Memory and aging in context. Psychological bulletin, 131(3), 383.

Hill, P. F., King, D. R., & Rugg, M. D. (2021). Age differences in retrieval-related reinstatement reflect age-related dedifferentiation at encoding. Cerebral Cortex, 31(1), 106-122.

Klein, A., Ghosh, S. S., Bao, F. S., Giard, J., Häme, Y., Stavsky, E., ... & Keshavan, A. (2017). Mindboggling morphometry of human brains. PLoS computational biology, 13(2), e1005350.

Koen, J. D., & Rugg, M. D. (2019). Neural dedifferentiation in the aging brain. Trends in cognitive sciences, 23(7), 547-559.

Koen, J. D., & Yonelinas, A. P. (2014). The effects of healthy aging, amnestic mild cognitive impairment, and Alzheimer's disease on recollection and familiarity: A meta-analytic review. Neuropsychology review, 24(3), 332-354.

Koutstaal, W., & Schacter, D. L. (1997). Gist-based false recognition of pictures in older and younger adults. Journal of memory and language, 37(4), 555-583.

Levine, B., Svoboda, E., Hay, J. F., Winocur, G., & Moscovitch, M. (2002). Aging and autobiographical memory: dissociating episodic from semantic retrieval. Psychology and aging, 17(4), 677.

Maass, A., Landau, S., Baker, S. L., Horng, A., Lockhart, S. N., La Joie, R., ... & Alzheimer's Disease Neuroimaging Initiative. (2017). Comparison of multiple tau-PET measures as biomarkers in aging and Alzheimer's disease. Neuroimage, 157, 448-463.

McDonough, I. M., Cervantes, S. N., Gray, S. J., & Gallo, D. A. (2014). Memory's aging echo: Age-related decline in neural reactivation of perceptual details during recollection. NeuroImage, 98, 346-358.

Nasreddine, Z. S., Phillips, N. A., Bédirian, V., Charbonneau, S., Whitehead, V., Collin, I., ... & Chertkow, H. (2005). The Montreal Cognitive Assessment, MoCA: a brief screening tool for mild cognitive impairment. Journal of the American Geriatrics Society, 53(4), 695-699.

Nastase, S. A., Gazzola, V., Hasson, U., & Keysers, C. (2019). Measuring shared responses across subjects using intersubject correlation. Social cognitive and affective neuroscience, 14(6), 667-685.

Nguyen, M., Vanderwal, T., & Hasson, U. (2019). Shared understanding of narratives is correlated with shared neural responses. NeuroImage, 184, 161-170.

Nyberg, L., Lövdén, M., Riklund, K., Lindenberger, U., & Bäckman, L. (2012). Memory aging and brain maintenance. Trends in cognitive sciences, 16(5), 292-305.

Palmqvist, S., Schöll, M., Strandberg, O., Mattsson, N., Stomrud, E., Zetterberg, H., ... & Hansson, O. (2017). Earliest accumulation of β-amyloid occurs within the default-mode network and concurrently affects brain connectivity. Nature communications, 8(1), 1214.

Pauley, C., Kobelt, M., Werkle-Bergner, M., & Sander, M. C. (2023). Age differences in neural distinctiveness during memory encoding, retrieval, and reinstatement. Cerebral Cortex, 33(16), 9489-9503.

Park, D. C., Polk, T. A., Park, R., Minear, M., Savage, A., & Smith, M. R. (2004). Aging reduces neural specialization in ventral visual cortex. Proceedings of the National Academy of Sciences, 101(35), 13091-13095.

Ranganath, C., & Ritchey, M. (2012). Two cortical systems for memory-guided behaviour. Nature reviews neuroscience, 13(10), 713-726.

Raz, N., Lindenberger, U., Rodrigue, K. M., Kennedy, K. M., Head, D., Williamson, A., ... & Acker, J. D. (2005). Regional brain changes in aging healthy adults: general trends, individual differences and modifiers. Cerebral cortex, 15(11), 1676-1689.

Reagh, Z. M., Delarazan, A. I., Garber, A., & Ranganath, C. (2020). Aging alters neural activity at event boundaries in the hippocampus and Posterior Medial network. Nature communications, 11(1), 3980.

Reagh, Z. M., & Ranganath, C. (2023). Flexible reuse of cortico-hippocampal representations during encoding and recall of naturalistic events. Nature Communications, 14(1), 1279.

Ritchey, M., & Cooper, R. A. (2020). Deconstructing the posterior medial episodic network. Trends in cognitive sciences, 24(6), 451-465.

Robin, J. (2018). Spatial scaffold effects in event memory and imagination. Wiley Interdisciplinary Reviews: Cognitive Science, 9(4), e1462.

Rugg, M. D., Johnson, J. D., Park, H., & Uncapher, M. R. (2008). Encoding-retrieval overlap in human episodic memory: a functional neuroimaging perspective. Progress in brain research, 169, 339-352.

Rugg, M. D., & Vilberg, K. L. (2013). Brain networks underlying episodic memory retrieval. Current opinion in neurobiology, 23(2), 255-260.

Rugg, M. D., & Srokova, S. (2024). Effects of age on neural reinstatement during memory retrieval. Reference Module in Neuroscience and Biobehavioral Psychology.

Schacter, D. L., Addis, D. R., Hassabis, D., Martin, V. C., Spreng, R. N., & Szpunar, K. K. (2012). The future of memory: remembering, imagining, and the brain. Neuron, 76(4), 677-694.

Schacter, D. L., Koutstaal, W., & Norman, K. A. (1997). False memories and aging. Trends in cognitive sciences, 1(6), 229-236.

Simony, E., Honey, C. J., Chen, J., Lositsky, O., Yeshurun, Y., Wiesel, A., & Hasson, U. (2016). Dynamic reconfiguration of the default mode network during narrative comprehension. Nature communications, 7(1), 12141.

Sperling, R. A., LaViolette, P. S., O'Keefe, K., O'Brien, J., Rentz, D. M., Pihlajamaki, M., ... & Johnson, K. A. (2009). Amyloid deposition is associated with impaired default network function in older persons without dementia. Neuron, 63(2), 178-188.

Srokova, S., Hill, P. F., Koen, J. D., King, D. R., & Rugg, M. D. (2020). Neural differentiation is moderated by age in scene-selective, but not face-selective, cortical regions. ENeuro, 7(3).

St-Laurent, M., Abdi, H., Bondad, A., & Buchsbaum, B. R. (2014). Memory reactivation in healthy aging: evidence of stimulus-specific dedifferentiation. Journal of Neuroscience, 34(12), 4175-4186.

St-Laurent, M., Abdi, H., & Buchsbaum, B. R. (2015). Distributed patterns of reactivation predict vividness of recollection. Journal of Cognitive Neuroscience, 27(10), 2000-2018.

Stern, Y. (2012). Cognitive reserve in ageing and Alzheimer's disease. The Lancet Neurology, 11(11), 1006-1012.

Stern, Y., Arenaza-Urquijo, E. M., Bartrés-Faz, D., Belleville, S., Cantilon, M., Chetelat, G., ... & Reserve, Resilience and Protective Factors PIA Empirical Definitions and Conceptual

Frameworks Workgroup. (2020). Whitepaper: Defining and investigating cognitive reserve, brain reserve, and brain maintenance. Alzheimer's & Dementia, 16(9), 1305-1311.

Therriault, J., Pascoal, T. A., Lussier, F. Z., Tissot, C., Chamoun, M., Bezgin, G., ... & Rosa-Neto, P. (2022). Biomarker modeling of Alzheimer's disease using PET-based Braak staging. Nature aging, 2(6), 526-535.

Trelle, A. N., Carr, V. A., Guerin, S. A., Thieu, M. K., Jayakumar, M., Guo, W., ... & Wagner, A. D. (2020). Hippocampal and cortical mechanisms at retrieval explain variability in episodic remembering in older adults. elife, 9, e55335.

Tulving, E. (1993). What is episodic memory?. Current directions in psychological science, 2(3), 67-70.

Wagner, A. D., Shannon, B. J., Kahn, I., & Buckner, R. L. (2005). Parietal lobe contributions to episodic memory retrieval. Trends in cognitive sciences, 9(9), 445-453.

Wang, Q., Cagna, B., Chaminade, T., & Takerkart, S. (2020). Inter-subject pattern analysis: A straightforward and powerful scheme for group-level MVPA. NeuroImage, 204, 116205.

Weintraub, S., Besser, L., Dodge, H. H., Teylan, M., Ferris, S., Goldstein, F. C., ... & Morris, J. C. (2018). Version 3 of the Alzheimer Disease Centers' neuropsychological test battery in the Uniform Data Set (UDS). Alzheimer Disease & Associated Disorders, 32(1), 10-17.

Wing, E. A., Ritchey, M., & Cabeza, R. (2015). Reinstatement of individual past events revealed by the similarity of distributed activation patterns during encoding and retrieval. Journal of cognitive neuroscience, 27(4), 679-691.

Zacks, J. M., Speer, N. K., Swallow, K. M., Braver, T. S., & Reynolds, J. R. (2007). Event perception: a mind-brain perspective. Psychological bulletin, 133(2), 273.

Zadbood, A., Nastase, S., Chen, J., Norman, K. A., & Hasson, U. (2022). Neural representations of naturalistic events are updated as our understanding of the past changes. Elife, 11, e79045.

Zhang, Y., Brady, J. M., & Smith, S. M. (2001). An hmrf-em algorithm for partial volume segmentation of brain mri fmrib technical report tr01yz1. Brain: A Journal of Neurology. https://www. fmrib. ox. ac. uk/datasets/techrep/tr01yz1/tr01yz1. pdf.